

ROTUMA ISLAND, Fiji

WMO: 916500

Lat: 12.500S

Lon: 177.050E

Elev: 26

StdP: 101.01

Time Zone: 12.00 (PCF)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.8	24.3	19.8	14.5	26.1	20.6	15.3	26.4	8.4	27.1	7.8	25.5	1.9	90	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.4	31.7	27.0	31.3	26.8	31.0	26.6	27.8	30.4	27.5	30.2	27.2	30.1	2.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	29.8	26.7	22.3	29.6	26.2	21.7	29.2	89.0	30.5	87.4	30.4	86.2	30.4	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	6.7	5.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	28.1	28.2	28.1	28.1	27.7	27.5	27.3	26.9	27.2	27.3	27.7	28.0	
	DBStd	0.98	1.01	0.87	0.90	0.83	0.87	0.87	0.82	0.87	0.95	0.88	0.90	0.96	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	6450	561	509	560	544	549	524	535	525	517	538	531	557	
	CDD18.3	3408	303	276	302	294	291	274	277	267	267	279	281	299	
	CDH23.3	36569	3453	3153	3414	3312	3104	2833	2781	2555	2683	2893	3029	3358	
	CDH26.7	9901	1133	1024	1086	1039	831	650	570	458	552	682	823	1052	
Wind	WSAvg	2.0	2.1	1.8	1.9	1.7	1.6	2.1	2.3	2.3	2.6	2.0	2.1	2.0	
Precipitation	PrecAvg	3475	349	322	348	277	283	262	234	213	216	340	310	310	
	PrecMax	4378	954	752	1006	572	524	554	471	601	445	656	609	664	
	PrecMin	2450	28	111	97	61	32	18	38	24	48	124	106	27	
	PrecStd	482	155	122	182	119	118	127	119	125	96	142	109	142	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	32.0	32.0	31.8	31.4	30.8	30.8	30.6	30.6	30.9	31.3	31.8	
		MCWB	27.2	27.2	27.2	27.0	27.0	26.5	26.4	26.5	26.0	26.3	26.4	27.0	
	2%	DB	31.5	31.3	31.4	31.2	30.7	30.3	30.2	30.0	30.1	30.3	30.7	31.2	
		MCWB	26.9	26.9	26.9	26.8	26.7	26.3	26.0	26.0	25.9	26.1	26.3	26.7	
	5%	DB	31.1	30.9	30.9	30.7	30.3	29.9	29.6	29.4	29.7	29.9	30.4	30.8	
		MCWB	26.7	26.6	26.7	26.6	26.4	26.1	25.8	25.7	25.8	25.9	26.1	26.5	
	10%	DB	30.6	30.5	30.4	30.3	29.8	29.4	29.1	28.9	29.3	29.5	29.9	30.2	
		MCWB	26.6	26.5	26.5	26.4	26.2	25.8	25.5	25.4	25.6	25.7	26.0	26.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.1	28.0	28.0	27.8	28.0	27.2	27.2	27.1	27.0	27.2	27.7	27.9
			MCDB	30.8	30.9	30.8	30.6	30.1	29.8	29.8	29.9	29.5	29.9	30.0	30.5
2%		WB	27.6	27.3	27.4	27.2	27.3	26.9	26.6	26.6	26.5	26.7	27.2	27.3	
		MCDB	30.4	30.5	30.3	30.2	29.7	29.6	29.1	29.2	29.0	29.5	29.7	30.2	
5%		WB	27.2	27.0	27.1	27.0	27.0	26.4	26.2	26.1	26.2	26.2	26.7	27.0	
		MCDB	30.2	30.3	30.0	30.1	29.5	29.0	28.7	28.6	28.7	29.0	29.3	30.0	
10%		WB	26.9	26.7	26.7	26.7	26.6	26.1	25.8	25.7	25.9	26.0	26.3	26.6	
		MCDB	29.8	29.7	29.6	29.6	29.1	28.6	28.3	28.1	28.4	28.7	28.9	29.4	
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.4	4.4	4.3	4.1	3.7	3.7	3.7	3.6	4.1	4.2	4.4	
		MCDBR	5.2	5.1	5.1	4.9	4.7	4.3	4.4	4.6	4.1	4.7	4.8	5.2	
		MCWBR	2.6	2.5	2.5	2.5	2.4	2.3	2.2	2.3	2.0	2.4	2.5	2.7	
	5% WB	MCDBR	4.7	4.9	4.8	4.6	4.5	4.0	4.1	4.2	4.0	4.4	4.5	4.9	
		MCWBR	2.7	2.6	2.5	2.5	2.5	2.4	2.3	2.4	2.2	2.5	2.8	2.8	
Clear-Sky Solar Irradiance	taub	0.409	0.405	0.395	0.394	0.401	0.420	0.421	0.418	0.422	0.426	0.417	0.416		
	taud	2.559	2.576	2.608	2.613	2.583	2.497	2.494	2.504	2.484	2.488	2.525	2.534		
	Ebn at Noon	935	932	923	892	854	818	827	858	884	902	920	927		
	Edh at Noon	109	106	100	95	93	98	100	104	112	115	112	111		
All-Sky Solar Radiation	RadAvg	5.69	5.50	5.60	5.10	4.55	4.24	4.49	5.07	5.63	5.77	5.78	5.75		
	RadStd	0.72	0.57	0.55	0.49	0.37	0.27	0.25	0.26	0.29	0.34	0.44	0.34		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page